

1. Prerequisites

- Install **Visual Studio 2022** (latest version recommended)
 - Select workloads:
 - **Universal Windows Platform development** (for WinUI)
 - **.NET Multi-platform App UI development** (for MAUI)
 - Windows 10/11 SDK installed
 - Optional: Android/iOS SDKs if targeting multiple platforms with MAUI
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2. Creating a WinUI 3 Project (Windows Only)

1. Open Visual Studio
2. Click **Create a new project**
3. Search for **Blank App, Packaged (WinUI 3 in Desktop)**
4. Name your project (e.g., `MyWinUIApp`), choose location and solution name
5. Select target framework (default: .NET 7)
6. Click **Create**

Project Structure

- `MainWindow.xaml` - Main UI layout
- `MainWindow.xaml.cs` - Code-behind
- `App.xaml` - Global resources and app definition
- `App.xaml.cs` - Startup logic

Run the Project

- Set **x64** as target
 - Press **F5** to build and run
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3. Creating a .NET MAUI Windows Project (Cross-platform)

1. Open Visual Studio
2. Click **Create a new project**
3. Select **.NET MAUI App** template
4. Name your project (e.g., `MyMauiApp`)
5. Choose target frameworks (Windows + Android recommended)
6. Click **Create**

Project Structure

- `MainPage.xaml` - UI layout
- `MainPage.xaml.cs` - Code-behind logic
- `Platforms/Windows` - Windows-specific code

- `Platforms/Android` - Android-specific code
- `App.xaml` / `App.xaml.cs` - Shared app startup logic

Run the Project

- Select **Windows Machine** target for Windows
- Press **F5** to run
- For Android, select **Android Emulator** target and press **F5**

4. Adding UI Elements (XAML)

- Use `<Grid>`, `<StackPanel>`, `<Border>`, `<Button>`, `<TextBlock>`, `<Image>`
- Set properties like `Width`, `Height`, `Padding`, `Margin`, `HorizontalAlignment`, `VerticalAlignment`
- Example:

```
<Border Background="#f0f0f0" Padding="10" CornerRadius="8">
    <StackPanel HorizontalAlignment="Center" VerticalAlignment="Center">
        <TextBlock Text="Welcome!" FontSize="24" FontWeight="Bold"/>
        <Button Content="Click Me" Margin="0,10,0,0"/>
    </StackPanel>
</Border>
```

5. Tips

- Use **Grid** for complex layouts
- Use **StackPanel** for vertical or horizontal stacking
- Use **Border** or **Frame** for card-like UI elements
- Always set **HorizontalAlignment** and **VerticalAlignment** for proper placement
- Test Windows build before moving to cross-platform targets (MAUI)

6. Running & Debugging

- Use **F5** to run and debug
- Use **Breakpoints** in XAML.cs code-behind for debugging logic
- Monitor output window for errors
- For MAUI, use different platform targets to test cross-platform UI behavior